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Intelligent Arabic Legal Advisor

**(Graduation Project 2) prepared to obtain the degree of B.Sc. in the Department of
Artificial Intelligence Engineering and Data Science**

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Supervisor certification

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Abstract

This project presents the design and implementation of an Intelligent Arabic Legal Advisor based on a Retrieval-Augmented Generation (RAG) architecture, specifically tailored for the Syrian legal framework. The system aims to move beyond traditional keyword-based legal search toward a context-aware and reasoning-driven platform capable of analyzing complex legal scenarios and providing evidence-based legal advice.

The system is built on a multi-stage pipeline designed for high accuracy and reliability. The retrieval layer employs a hybrid search strategy that combines semantic vector embeddings using the multilingual-e5-large model with lexical retrieval through the BM25 algorithm. To further enhance precision, an advanced neural reranking model (BAAI/bge-reranker-base) is integrated to reorder candidate legal articles based on deep semantic relevance.

Additionally, a weighted fusion mechanism is applied to balance semantic and lexical scores, improving retrieval robustness. The system also incorporates several reliability mechanisms, including query normalization for Arabic text, fallback strategies for query refinement, duplicate removal at the article level, and threshold-based filtering of reranked results.

The generation layer utilizes Large Language Models (LLMs) such as Llama 3 and Gemini via the OpenRouter API to perform legal reasoning and generate structured responses. The system ensures attribution by linking each conclusion to its corresponding legal source. This project demonstrates a scalable and modular Legal-AI solution that effectively bridges the gap between raw legal texts and practical legal consultation.

الملخص

يقدم هذا المشروع تصميم وتنفيذ نظام مستشار قانوني عربي ذكي يعتمد على بنية الاسترجاع المعزز بالتوليد (RAG) ، والمصمم خصيصاً ليتوافق مع الإطار القانوني السوري. يهدف النظام إلى تجاوز أساليب البحث التقليدية القائمة على الكلمات المفتاحية، والانتقال إلى منصة تحليلية تعتمد على فهم السياق والاستدلال، قادرة على تحليل الحالات القانونية المعقدة وتقديم استشارات قانونية موثقة.

تم بناء النظام على خط معالجة متعدد المراحل مصمم لتحقيق دقة وموثوقية عالية. تعتمد طبقة الاسترجاع على استراتيجية بحث هجينة تجمع بين التضمينات المتجهة الدلالية باستخدام نموذج multilingual-e5-large، والبحث المعجمي باستخدام خوارزمية BM25. ولتعزيز الدقة بشكل أكبر، تم دمج نموذج إعادة ترتيب عصبي متقدم (BAAI/bge-reranker-base) لإعادة ترتيب المواد القانونية بناءً على الصلة الدلالية العميقة.

كما تم تطبيق آلية دمج موزونة بين البحث الدلالي والمعجمي لتحسين أداء الاسترجاع. ويتضمن النظام أيضاً عدة آليات لتعزيز الموثوقية، مثل تطبيع الاستعلامات العربية، واستخدام آليات احتياطية في إعادة صياغة الأسئلة، وإزالة التكرار على مستوى المواد القانونية، بالإضافة إلى تصفية النتائج باستخدام عتبة ثقة محددة.

تعتمد طبقة التوليد على نماذج لغوية كبيرة مثل Llama 3 و Gemini عبر واجهة OpenRouter ، حيث تقوم بعمليات الاستدلال القانوني وتوليد إجابات منظمة. ويضمن النظام توثيق الإجابات من خلال ربط كل استنتاج بمصدره القانوني. يبرز هذا المشروع حلاً متكاملاً وقابلاً للتوسع في مجال الذكاء الاصطناعي القانوني، يسد الفجوة بين النصوص القانونية الخام والتطبيق العملي للاستشارات القانونية.

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List of Abbreviations

(Full Description)	(Abbreviation)
Retrieval-Augmented Generation	RAG
Large Language Model	LLM
Natural Language Processing	NLP
Best Matching 25	BM25
Multilingual Embedding Model	E5-Large
Bigai Generalized Embedding Reranker	BGE Reranker
Optical Character Recognition	OCR
(RAG Assessment) The Evaluation Framework for System Quality used in Chapter 5	RAGAS
Vector Database used for indexing legal chunks	ChromaDB
(GPU RTX 3050) Compute Unified Device Architecture	CUDA

List of Symbols

Reference in Research	(Description)	(Symbol)
Chapters 4 & 5	Represents the User Query input into the system.	Q
Chapters 4 & 5	Represents the retrieved Legal Article or document.	D
Equation 2 (Chapter 5)	Cosine Similarity used to measure the relevance between the query and the answer.	$\cos(\theta)$
Equation 3 (Chapter 5)	The number of top-ranked results retrieved (Top-k Results).	k
Equation 3 (Chapter 5)	Indicator function that equals 1 if the retrieved segment is relevant.	v_i
Equations 1 & 2	The total number of Claims or questions used in the evaluation.	n
Chapters 3 & 4	The score of the Lexical Search using the BM25 algorithm.	$S\{Lexical\}$
Chapters 3 & 4	The score of the Semantic Search using Vector Embeddings.	$S\{Semantic\}$

Chapter 1

Introduction

1.1 Introduction

This project presents the development of an Intelligent Arabic Legal Consultant System based on Retrieval-Augmented Generation (RAG), semantic search, and advanced natural language reasoning. The system is designed to provide automated, evidence-based legal consultations by analyzing Syrian legislative texts and judicial precedents without requiring manual document search.

The system is implemented as a modular multi-stage pipeline using Python, enabling efficient data processing, indexing, retrieval, and response generation. The retrieval layer is built on a Hybrid Search mechanism that combines semantic vector embeddings using the multilingual-e5-large model with traditional lexical retrieval through the BM25 algorithm. To further improve retrieval accuracy, the system integrates a neural reranking model (BAAI/bge-reranker-base), which reorders candidate legal articles based on deep semantic relevance.

In addition, a weighted fusion strategy is applied to combine semantic and lexical retrieval scores, ensuring balanced performance between contextual understanding and exact term matching. The system also incorporates several robustness mechanisms, including Arabic query normalization, query refinement fallback strategies, article-level deduplication, and threshold-based filtering of reranked results. These enhancements significantly improve retrieval precision and reduce irrelevant or noisy outputs.

Users interact with the system through natural language queries in Arabic, describing legal situations in a narrative form. The system processes the query, retrieves the most relevant legal articles, applies reranking, and generates a structured legal response using Large Language Models (LLMs) such as Llama 3 and Gemini via the OpenRouter API. Each generated response is grounded in retrieved legal texts, ensuring factual consistency and source attribution.

This project demonstrates a scalable and practical Legal-AI solution that bridges the gap between raw legal documents and intelligent legal consultation systems, making it suitable for academic research and real-world legal assistance applications.

1.2 Problem Statement

Despite the digital transformation in various sectors, many legal research tasks in the Syrian judicial context still rely on manual document review or keyword-based search systems that lack semantic understanding and contextual autonomy. These traditional approaches are highly inefficient for processing the vast and

complex structures of legislative texts and judicial precedents. They do not scale well in professional or academic environments where precise, evidence-backed legal answers are required quickly.

There is a critical need for Intelligent Legal Systems that can perceive the semantic nuances of Arabic legal language, make independent reasoning decisions over unstructured data, and execute accurate information retrieval reliably.

This project addresses this problem by developing an Autonomous Legal Consultant System based on Retrieval-Augmented Generation (RAG). The system is designed for vision-independent, semantic-based legal sorting and reasoning, initiated through natural language queries to provide a modular and scalable solution for modern legal challenges.

Additionally, traditional systems lack mechanisms for result validation and ranking optimization, leading to irrelevant or misleading outputs. This highlights the need for advanced reranking and filtering techniques in modern Legal-AI systems.

1.3 Project Objectives

The primary objective of this project is to bridge the gap between complex legal archives and intelligent automated assistance. To achieve this, the following specific objectives have been established:

- To Develop a Semantic Retrieval Pipeline: Building a system capable of understanding the legal context of user queries rather than relying on exact keyword matching, specifically for Syrian Law.
- To Implement an Arabic-specific RAG Architecture: Integrating advanced embedding models (like multilingual-e5-large) and vector databases (ChromaDB) to handle the nuances of Arabic legal terminology.
- To Automate Legal Reasoning: Leveraging Large Language Models (LLMs) to synthesize accurate, documented, and concise legal advice based on retrieved articles and judicial precedents.
- To Ensure System Modularity and Scalability: Utilizing frameworks like LangChain or clustering to create a flexible architecture that can be easily expanded to include other branches of law or more complex datasets in the future.

1.4 Research Contribution

This research provides several key contributions to the field of Arabic Legal Informatics, specifically tailored for the Syrian judicial context:

- **Specialized Syrian Legal RAG Framework:** Development of a Retrieval-Augmented Generation (RAG) system specifically designed to handle the complexities of Syrian legislations (Penal, Civil, and Personal Status Laws), providing a context-aware alternative to traditional keyword-based search.
- **Structural Article-Based Chunking:** Introduction of a novel "Structural Chunking" methodology that utilizes regular expressions (Regex) to segment legal texts by individual articles rather than fixed character lengths. This preserves the semantic integrity and legal logic of each provision, ensuring higher reasoning accuracy.
- **Multi-Stage Retrieval Optimization:** Implementation of a high-precision pipeline that integrates hybrid search (Semantic E5-Large + Lexical BM25) with a neural reranking layer (BGE Reranker). This combination effectively addresses the nuances of formal Arabic legal terminology and improves the signal-to-noise ratio in retrieved results.
- **Evidence-Based Grounding Mechanism:** Establishing a hallucination-resistant architecture that strictly anchors Large Language Model (LLM) responses to specific Syrian legal articles, ensuring that every piece of advice is documented and verifiable

1.5 Intended Beneficiaries

The "Intelligent Arabic Legal Advisor" is designed to serve a diverse range of stakeholders within the Syrian legal ecosystem. By bridging the gap between raw, unstructured legislative texts and intelligent automated consultations, the system provides significant value to the following groups:

- **Legal Professionals (Lawyers and Legal Consultants):** The system acts as a high-speed research assistant, enabling lawyers to bypass the manual review of thousands of legal articles. By providing instant access to specific provisions from the **Syrian Penal, Civil, and Personal Status laws**, it allows professionals to focus on strategic case analysis rather than time-consuming documentation.

- **The General Public and Citizens:** A primary goal of this system is the **democratization of legal knowledge**. It provides citizens with an accessible platform to understand their rights and obligations through natural language queries, translating complex judicial terminology into actionable legal advice.
- **Judicial and Governmental Institutions:** The project serves as a model for the **digital transformation** of the Syrian judicial sector. It provides a scalable foundation for automating court ruling analysis and administrative legal tracking, which could eventually lead to faster judicial proceedings and reduced clerical errors.
- **Academic and Research Institutions:** As an expert system built on advanced RAG architecture, it serves as a valuable **educational and experimental platform** for students and researchers in the fields of Artificial Intelligence and Legal Informatics. It fills a significant research gap by providing a specialized framework for Arabic-language legal reasoning.
- **Small and Medium Enterprises (SMEs):** SMEs often lack the resources for full-time legal counsel. This system provides them with a preliminary tool to verify contract obligations and labor regulations, ensuring basic legal compliance without initial high costs.

1.6 Project Scope

The scope of this project encompasses a diverse range of **Syrian Legislations**, extending beyond the Civil Law to include multiple key legal branches and selected judicial precedents from the **Court of Cassation**. The system is optimized to process complex legal Arabic and provide reasoning across integrated datasets. The current implementation covers the end-to-end RAG pipeline: from sophisticated document chunking and semantic indexing in **ChromaDB** to automated response synthesis using advanced LLMs. While the system supports a multi-law framework, the integration of real-time judicial database synchronization and specialized criminal forensics remains outside the current scope and is reserved for future enhancements.

The system also focuses on improving retrieval quality through ranking optimization and filtering techniques.

1.7 Challenges in Legal-AI Integration

Developing an intelligent legal assistant requires seamless integration of semantic retrieval, natural language reasoning, and contextual accuracy. Coordinating high-dimensional vector search with large-scale language models (LLMs) while maintaining the integrity of Syrian legal provisions presents a significant engineering challenge. Introducing Arabic-specific nuances further increases system complexity, requiring precise document chunking and sophisticated prompting strategies to ensure that the generated advice is both legally sound and contextually relevant. Another key challenge is ensuring that reranking models operate efficiently within limited computational resources.

1.8 Importance of the Project

This project demonstrates how advanced Legal-AI systems can be developed in an accessible and modular manner without relying on proprietary, expensive legal platforms or closed-source intelligence frameworks. It provides a practical framework for understanding the interaction between Information Retrieval (IR), Vector Databases, and Generative AI within a unified legal consultation pipeline.

The system serves as a valuable educational and experimental platform for the Syrian judicial sector and establishes a scalable foundation for future extensions, including automated court ruling analysis, real-time legislative tracking, and enhanced multi-lingual legal support.

1.9 Report Structure

This project report is organized into six main chapters, each addressing a specific phase of the research and development process:

- **Chapter One: Introduction:** Provides an overview of the project, the problem statement regarding the accessibility of Syrian law, the proposed AI-based solution, and the specific objectives this research aims to achieve.
- **Chapter Two: Fundamental Concepts and Literature Review:** Establishes the theoretical foundation by discussing Natural Language Processing (NLP) and the evolution of Large Language Models (LLMs). It explores the mechanics of Retrieval-Augmented Generation (RAG) and semantic search, while also reviewing existing AI applications in the legal domain to highlight the gap this project fills within the Syrian context.

- **Chapter Three: Dataset and Preprocessing:** Focuses on the data engineering phase, covering the acquisition of Syrian legal texts and the comprehensive preprocessing pipeline. This includes OCR extraction, text normalization, and the "Structural Chunking" strategy used to transform raw laws into a machine-readable format suitable for AI processing.
- **Chapter Four: System Design and Implementation:** Describes the complete technical architecture of the Legal Advisor system. It details the integration of vector databases (ChromaDB), the development of the hybrid retrieval and reranking modules, the hardware optimization for local execution, and the design of the user interface.
- **Chapter Five: Results and Evaluation:** Presents the experimental results and performance analysis. It evaluates the system using quantitative metrics (such as Faithfulness and Relevancy) and qualitative case studies to assess the accuracy of the legal advice generated in real-world scenarios.
- **Chapter Six: Conclusion and Future Work:** Summarizes the project's primary contributions to the field of Syrian legal informatics. It also identifies current limitations and suggests future directions for expanding the system's knowledge base and reasoning capabilities.

Chapter 2

Fundamental Concepts and Literature Review

2.1 Introduction

This chapter provides a foundational understanding of the Smart Arabic Legal Consultant system by examining key concepts and reviewing relevant studies in the field. It explores essential terminologies, theoretical foundations, and the principles underlying the AI-based techniques utilized in the system. The purpose of this chapter is to present an overview of artificial intelligence applications in legal search systems, highlighting the role of Retrieval-Augmented Generation (RAG), semantic search, and Arabic Natural Language Processing (NLP). Additionally, the chapter reviews existing solutions and academic contributions to identify current challenges and advancements, thereby supporting and guiding the development of the proposed legal framework.

2.2 Fundamental Concepts

This section provides an in-depth explanation of the core technical concepts underpinning the system. It introduces the frameworks that enable intelligent retrieval and organization of legal materials.

- **Artificial Intelligence (AI) and Machine Learning (ML):** AI refers to systems designed to perform tasks requiring human intelligence, such as legal reasoning. In this project, ML models are utilized to automatically extract semantic features from Syrian legal codes (Penal and Personal Status Laws) to support intelligent indexing.
- **Natural Language Processing (NLP):** NLP is a subfield of AI focusing on the interaction between computers and human language. Within this project, NLP techniques are employed to process complex Arabic legal materials. Core processes include **Tokenization**, **Semantic Embedding generation**, and **Arabic Reshaping**, which allow the system to interpret the context of legal queries rather than relying on simple keyword matching.
- **Retrieval-Augmented Generation (RAG):** RAG is an architectural framework that enhances Large Language Models (LLMs) by integrating an external, authoritative knowledge base. Unlike standalone generative models that rely solely on internal parameters, RAG forces the model to retrieve specific evidence—in this case, Syrian legal articles—before synthesizing a response. This architecture is governed by the principle of grounding, which ensures that the output is logically anchored to the

retrieved context, thereby minimizing "hallucinations" and ensuring factual integrity in legal reasoning.

- **Re-ranking in RAG Systems:** Re-ranking is a critical stage in modern RAG pipelines, where initially retrieved documents are re-ordered based on their semantic relevance to the user query. In this project, a transformer-based reranker model (BGE Reranker) is utilized to evaluate query-document pairs and assign relevance scores. This step significantly improves precision by filtering out weak candidates and ensuring that only the most contextually relevant legal articles are passed to the generation stage.
- **Information Retrieval (IR) and Semantic Search:** Unlike traditional search, semantic search utilizes vector-based retrieval. Embedding models (such as BERT or E5) map legal data into a shared semantic space, enabling retrieval based on conceptual similarity (e.g., matching "penalty" with "sanction").
- **Hybrid Retrieval with Score Fusion:** The system employs a hybrid retrieval strategy that combines semantic vector search with traditional BM25 keyword matching. Instead of using simple merging techniques, a weighted score fusion mechanism is applied, where vector similarity and lexical relevance are combined using predefined weights. This approach ensures balanced retrieval performance, capturing both deep semantic meaning and exact legal terminology.
- **Arabic Natural Language Processing (ANLP) in Law:** Legal Arabic is characterized by highly formal structure and complex morphology. The system employs NLP techniques such as **Tokenization** and **Semantic Embedding** to ensure that legal terms (e.g., "تعزير", "جنحة", "جناية") are understood within their specific judicial context rather than as general vocabulary.
- **Retrieval-Augmented Generation (RAG) Architecture:** Unlike standard AI models that may "hallucinate" or invent legal articles, RAG acts as a bridge between a Large Language Model (LLM) and the **Syrian Penal and Personal Status Codes**. It ensures that every response is grounded in a specific, retrieved legal text, providing the "Faithfulness" required for legal applications.

- **Vector Embeddings and Semantic Space :** At the core of semantic search is the transformation of textual data into high-dimensional numerical vectors. Using models like Multilingual-E5-Large, legal provisions are mapped into a 1024-dimensional vector space. The system calculates the relevance between a user query () and a legal document () using Cosine Similarity, which measures the angular distance between their respective vectors:

$$similarity = \cos(\theta) = \frac{Q \cdot D}{\|Q\| \|D\|}$$

Equation 1similarity matrix

This allows the system to identify conceptual relationships, such as linking "divorce procedures" with the formal term "أركان الطلاق", even without exact keyword matches.

- **Structural Chunking (Article-Based):** A core concept in this project is "Article-based Chunking." Instead of splitting text by size, the system identifies the boundaries of each **Legal Article (المادة)**. This preserves the integrity of the legal provision and ensures that the AI receives a complete, non-fragmented context for reasoning.
- **Hybrid Retrieval and Lexical Matching (BM25)** To balance semantic understanding with precise term matching, the system utilizes **BM25 (Best Matching 25)** for lexical retrieval. BM25 is a probabilistic ranking function that evaluates the relevance of documents based on term frequency and inverse document frequency. This is crucial for legal systems where specific terms (e.g., "جناية" vs. "جنحة") have distinct legal consequences. The final retrieval score is often a **Weighted Fusion** of the semantic and lexical scores to ensure a robust search pipeline
- **Neural Reranking (Cross-Encoders)** While initial retrieval identifies broad candidates, a **Neural Reranking** layer using a **Cross-Encoder** (like BGE Reranker) is applied for high-precision sorting. Unlike Bi-Encoders that process queries and documents independently, Cross-Encoders process the query-document pair simultaneously, capturing deep semantic interactions. This stage acts as a "neural filter" that eliminates irrelevant chunks that may have high keyword overlap but low legal relevance.

2.3 Literature Review

This review focuses on the current state of Arabic AI systems and studies that directly influence the development of our legal consultant.

2.3.1 Comparative Analysis of Knowledge Systems

We evaluate existing platforms to highlight the necessity of a specialized Syrian Legal RAG system:

- **Google Scholar & Semantic Scholar:** While powerful for academic research, they lack the interactive "consultant" persona and are not optimized for the specific hierarchy of Syrian legislations.
- **Arabic Legal RAG Datasets (2024):** Recent datasets on platforms like Hugging Face (e.g., Egyptian Legal RAG) prove that RAG is the most effective architecture for legal QA, but they lack support for the Syrian jurisdiction.

2.3.2 Analysis of Recent Arabic RAG Studies (2024-2025)

Based on recent research, the following studies have informed the technical direction of our system:

Study Title (Year)	Core Focus	Performance / Findings
Evaluating RAG Pipelines for Arabic (2025)	Lexical IR with 88k entries	GPT-4 achieved Accuracy≈0.82 ; highlighted the weakness of traditional embeddings in Arabic.
Exploring RAG in Arabic (2024)	Diverse Arabic texts & dialects	Showed significant improvement in answer quality compared to standalone LLMs.
Optimizing RAG Pipelines (2025)	Systematic Analysis of Arabic Docs	Proven that Semantic Chunking provides the best performance for long documents.

Study Title (Year)	Core Focus	Performance / Findings
Arabic Legal RAG Dataset (HF)	Egyptian Legislations	Confirmed that RAG is essential for factual legal accuracy and source attribution.

Table 1Recent RAG studies

2.3.3 Analysis of Embedding Models and Supporting Literature

Embedding Model	Related Study / Research	Findings in Arabic Context	Effectiveness for Syrian Law
BGE-M3	<i>Optimizing RAG Pipelines for Arabic (2025)</i>	Proven to be the best for Semantic Chunking . It excels in capturing the relationship between long legal sentences.	Superior: Best for handling the complex phrasing of the Syrian Penal Code.
AraBERT / Arabic-BERT	<i>Evaluating RAG Pipelines for Arabic (2025)</i>	High performance in lexical retrieval. Achieved an F1-score of ≈ 0.90 when combined with GPT-4.	High: Very strong at understanding formal "Modern Standard Arabic" used in legislations.
E5 (Multilingual)	<i>Evaluating RAG Pipelines (2025)</i>	One of the most stable models for cross-lingual alignment . Recommended for Arabic Lexical IR.	Very Good: Excellent for matching a user's everyday story to

Embedding Model	Related Study / Research	Findings in Arabic Context	Effectiveness for Syrian Law
			the formal text of the law.

Table 2 Embedding models literature

As shown in the table, the selection of models for the Smart Arabic Legal Consultant system was carefully designed based on recent research. While embedding models such as E5 are used for semantic retrieval, the system incorporates an advanced reranking stage using the BGE Reranker model. This combination allows the system to first retrieve a broad set of candidate legal texts and then refine them to the most relevant subset. Additionally, recent improvements such as query normalization, hybrid score fusion, and threshold-based filtering further enhance the robustness and accuracy of the retrieval pipeline, ensuring reliable legal responses

2.4 Conclusion

The literature confirms that the most significant challenge in Arabic Legal-AI is the accurate retrieval of specialized text. By adopting Article-based Chunking and a RAG-based framework, the Smart Arabic Legal Consultant addresses the gaps identified in current research. This foundation allows the system to move beyond simple search, providing a simulated "human" legal advisor that is both accurate and contextually aware.

Chapter 3

Dataset and Preprocessing

3.1 Introduction

The effectiveness of any Retrieval-Augmented Generation (RAG) system is fundamentally dependent not only on data quality but also on the design of its retrieval and reasoning pipeline. This chapter presents a comprehensive overview of the data lifecycle and system processing stages in the Intelligent Arabic Legal Advisor. It covers dataset acquisition, preprocessing, semantic indexing, and the integration of hybrid retrieval and reranking mechanisms.

The proposed system extends beyond traditional RAG implementations by incorporating multiple optimization layers, including query refinement, Arabic text normalization, hybrid retrieval (vector + lexical), and relevance-based reranking. These enhancements ensure that the system can retrieve highly accurate and contextually relevant legal provisions in response to user queries written in natural Arabic.

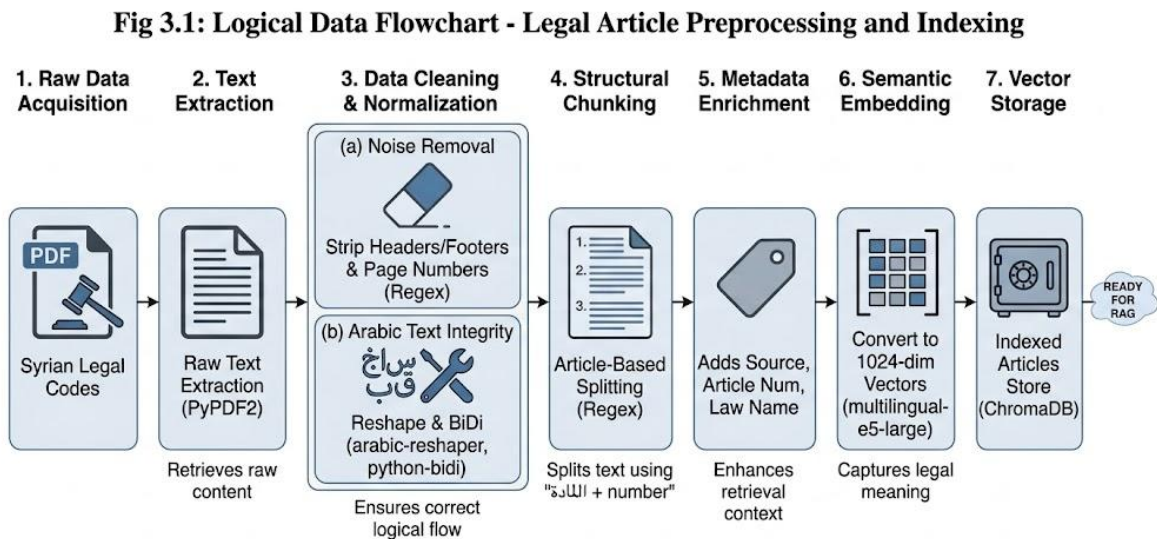


Fig. 3.1: Logical flowchart illustrating the comprehensive pipeline that transforms raw Syrian legal PDF documents into structured, context-rich, and searchable vector embeddings for the RAG system.

Figure 1Data flowchart

3.2 Dataset Acquisition:

The dataset comprises primary Syrian legal sources, selected to cover the most frequent legal inquiries in civil and criminal domains.

Data Sources

The system indexes three pillars of the Syrian legal framework:

- **Syrian Penal Code (Legislative Decree No. 148 of 1949):** Focused on criminal acts, sanctions, and felony/misdemeanor classifications.
- **Syrian Personal Status Law (Law No. 59 of 1953 and its amendments):** Covering family law, marriage, divorce, and inheritance.
- **Syrian Civil Law (Civil Code):** Regulating civil rights, contracts, and obligations.

Data Characteristics

The raw data was collected in PDF format. These documents present unique challenges for NLP, including:

- **Formal Legal Arabic:** High use of archaic and specialized judicial vocabulary.
- **Complex Formatting:** Multi-column layouts, footnotes, and recurring administrative headers/footers.
- **Document Noise:** OCR artifacts and page numbering that disrupt text continuity.

3.3 Preprocessing Pipeline

A robust preprocessing pipeline was developed using Python to transform raw legal documents into structured, high-quality data suitable for retrieval and reasoning. This pipeline includes multiple stages: text cleaning, normalization, structural chunking, and metadata enrichment.

In addition to standard preprocessing, the system incorporates Arabic-specific normalization techniques to improve retrieval performance. These include unifying different forms of Arabic characters (e.g., "ا", "أ", "إ") and removing diacritics to ensure consistency between user queries and indexed legal texts.

3.3.1 Text Cleaning and Normalization

To ensure the embedding model focuses only on legal content, we implemented several cleaning steps:

- **Noise Removal:** Automated scripts to strip recurring headers (e.g., "UNESCO Disclaimer" in certain versions of the Penal Code) and footers.
- **Number Standardization:** Page numbers in both Western (1, 2, 3) and Arabic-Indic (١, ٢, ٣) numerals were removed to prevent indexing irrelevant numeric data.

- **Arabic Text Integrity:** Using arabic-reshaper and python-bidi, we ensured that the text remains logically correct for the embedding model while being visually accurate for the UI display.

3.3.2 Structural Article-Based Chunking

Unlike standard RAG systems that use fixed-size character chunking (e.g., every 500 characters), this project utilizes **Structural Article-Based Chunking**.

- **Mechanism:** Using Regular Expressions (Regex), the system identifies the pattern "المادة" followed by a number.
- **Rationale:** Legal provisions are self-contained units. Splitting a legal article in the middle would destroy its semantic meaning. By chunking by article, we ensure that the retriever always returns a complete legal rule.

3.4 Metadata Enrichment

Each chunk is not just stored as text; it is enriched with **Metadata** to improve retrieval and source attribution:

- **law_name:** (e.g., Penal Code).
- **article_number:** (e.g., 155).
- **source_file:** The filename for traceability.

3.5 Vector Database and Embedding

The processed legal chunks are transformed into 1024-dimensional dense vector representations using the Multilingual-E5-Large embedding model. These embeddings capture semantic meaning and allow the system to perform similarity-based retrieval.

The vectors are stored in ChromaDB, which enables efficient and scalable semantic search. This vector-based retrieval is complemented by a lexical search component (BM25), forming a hybrid retrieval architecture that balances semantic understanding and exact keyword matching.

3.6 Hybrid Retrieval and Query Processing

To enhance retrieval performance, the system employs a hybrid retrieval strategy that combines semantic vector search with BM25 keyword-based search. Instead of relying on a single retrieval method, the system merges results using a weighted score fusion mechanism, ensuring that both semantic similarity and lexical matching contribute to the final ranking.

Additionally, the system applies query preprocessing techniques before retrieval, including query refinement using a language model and Arabic text normalization. These steps improve the quality of the input query and significantly increase retrieval accuracy.

3.7 Reranking and Filtering

After retrieval, the system applies a reranking stage using the BGE Reranker (bge-reranker-base), a transformer-based cross-encoder model. This model evaluates each query-document pair and assigns a relevance score, allowing the system to reorder the retrieved results based on contextual accuracy.

To improve reliability, a threshold-based filtering mechanism is applied, where only documents exceeding a predefined relevance score are considered. This reduces noise and prevents irrelevant legal articles from reaching the generation stage.

3.8 Response Safety and Generation

The final stage of the pipeline involves response generation using a Large Language Model (LLM). The system employs strict prompt engineering to ensure that the model does not hallucinate or generate unsupported legal claims.

Several safety mechanisms are implemented, including:

Restricting the model to use only retrieved legal texts

Detecting weak or insufficient context and returning a fallback response

Limiting the number of input articles to the most relevant subset

These measures ensure that the system produces accurate, trustworthy, and legally grounded responses.

3.9 Expert System Persona and Response Logic

To elevate the system from a basic document retriever to an intelligent legal assistant, a specialized **Legal Persona** was engineered.

3.9.1 System Prompt Engineering

The system utilizes advanced prompt engineering to guide the Large Language Model (LLM) to act as a "**Syrian Legal Consultant**". This involves:

- **Contextual Fact Analysis:** Evaluating the user's narrative to identify legal triggers.
- **Legal Qualification (Takyif):** Linking specific actions to legal definitions.
- **Integrated Reasoning:** Synthesizing multiple legal articles into a cohesive advisory response.

3.9.2 Structural Response Framework

Every response follows a professional legal methodology:

1. **Fact Comprehension:** Analyzing the user's situation.
2. **Legal Adaptation:** How the law views the facts.
3. **Applicable Provisions:** Natural integration of relevant articles.
4. **Actionable Recommendations:** Practical next steps for the user.

3.10 Preprocessing Pipeline Summary

Stage	Action	Tool / Library
Extraction	Converting scanned PDF pages into text using OCR	PyMuPDF (fitz) + Tesseract OCR
Cleaning	Removing OCR noise, long numbers, and non-Arabic characters	Custom Python Regex
Normalization	Standardizing Arabic characters (e.g., ا → أ)	Custom Python Functions
Text Refinement	Fixing OCR errors (e.g., "المادة" → "المادة") and formatting issues	Regex-based correction
Chunking	Splitting text into legal articles based on "المادة" pattern	Regex Patterns
Dataset Construction	Structuring data into JSON format with metadata	Python (JSON, re)
Retrieval	Hybrid Search (Vector + BM25)	ChromaDB + BM25
Reranking	Relevance-based ranking of retrieved chunks	BGE Reranker (BAAI/bge-reranker-base)
Query Processing	Query refinement and normalization before retrieval	LLM + Regex

Vectorization	Converting text into densevector embeddings	Multilingual-E5-Large
Storage	Vector indexing and persistence	ChromaDB

Table 3 Preprocessing Pipeline Summary

3.11 Chapter Summary

This chapter presented a comprehensive overview of the data lifecycle and system architecture. Beyond traditional preprocessing, the system integrates hybrid retrieval, query optimization, reranking, and safety mechanisms. These enhancements significantly improve retrieval accuracy and response reliability, making.

Chapter 4

System Design and Implementation

4.1 Introduction

system. It provides a detailed explanation of the system architecture, core components, and the interaction between different modules. The system is built using a modular pipeline approach that integrates data preprocessing, hybrid retrieval, reranking, and large language model (LLM) reasoning. The objective of this design is to ensure high accuracy, scalability, and reliability when handling complex Arabic legal queries.

4.2 System Architecture Overview

The system follows a Retrieval-Augmented Generation (RAG) architecture composed of multiple interconnected layers. These layers include data ingestion, preprocessing, indexing, retrieval, reranking, and response generation. Each layer performs a specific task, ensuring a clear separation of concerns and improving maintainability.

The overall pipeline can be summarized as follows:

OCR Extraction → Cleaning & Normalization → Chunking → Dataset Construction → Vector Indexing → Hybrid Retrieval → Reranking → LLM Generation

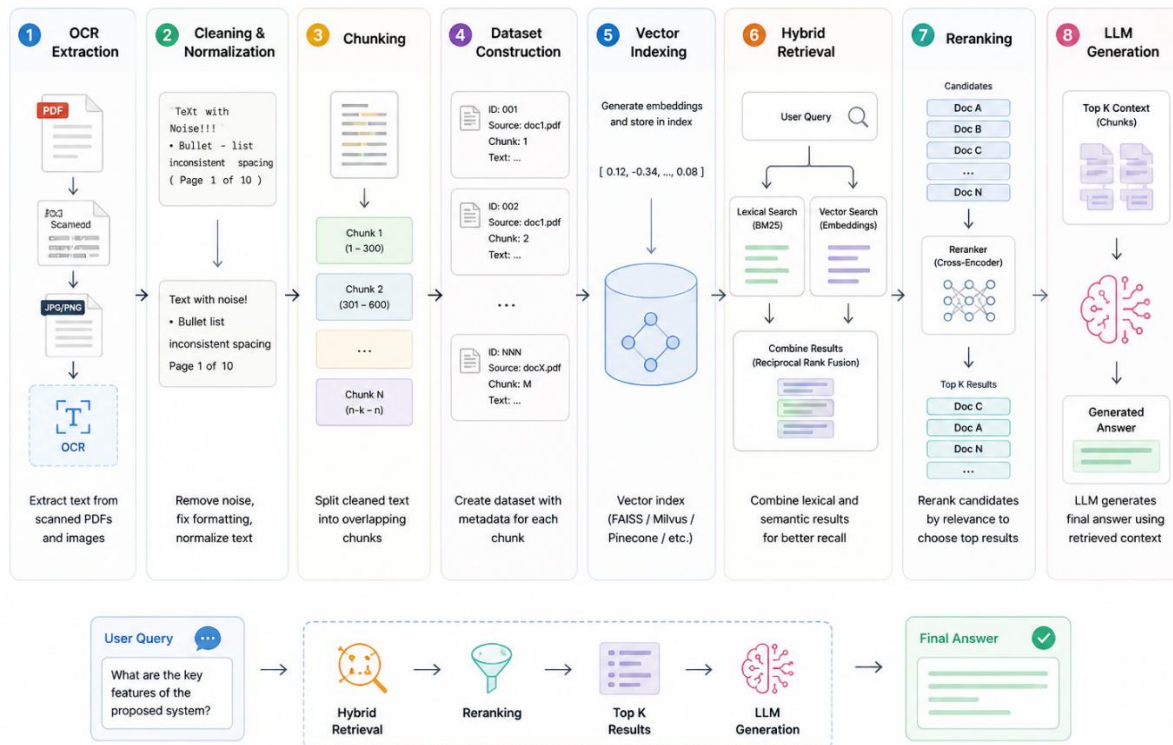


Figure 2 System architecture

4.3 Technical Specifications

To ensure the robustness of the "Intelligent Arabic Legal Advisor," the following technical stack was utilized:

Component	Specification
Primary Language	Python (PyTorch, Transformers, LangChain)
Vector Database	ChromaDB (v2_clean)
Embedding Model	intfloat/multilingual-e5-large (1024-dim)
Reranker Model	BAAI/bge-reranker-v2-m3
LLM Provider	OpenRouter (Meta Llama 3.3 70B)
Hardware Optimization	CUDA-enabled (RTX 3050 Optimized)

Table 4 Project Technical Specifications

4.4 Data Processing Pipeline

- The data processing pipeline transforms raw legal documents into structured, machine-readable data. The system uses Optical Character Recognition (OCR) to extract text from scanned Syrian law PDFs.
- Normalization: Standardizing Arabic characters and removing OCR noise.
- Structural Chunking: Segmenting text into legal articles using regex-based pattern matching to preserve legal context.
- Dataset Construction: Creating a structured JSON dataset enriched with metadata such as article numbers and law names.

نموذج التضمين Embedding) (Model	محتوى المادة Article) (Content	نوع القانون Law) (Type	معرف المادة Metadata) (ID
E5-Large (1024-dim)	"يعاقب بالأشغال الشاقة..."	قانون العقوبات	SY-PC-155
E5-Large (1024-dim)	"يجب على الزوج..."	الأحوال الشخصية	SY-PS-20

Table 5 Sample of Structured Legal Data and Metadata Storage in ChromaDB

4.5 Hybrid Retrieval System

The system implements a hybrid search mechanism to balance contextual meaning and exact legal terminology:

- Semantic Search: Performed using dense vector embeddings generated by the multilingual-e5-large model stored in ChromaDB.
- Lexical Search: Performed using the BM25 algorithm for precise keyword matching.
- Score Fusion: A weighted fusion strategy is applied, giving higher importance to semantic retrieval to capture the nuances of legal language

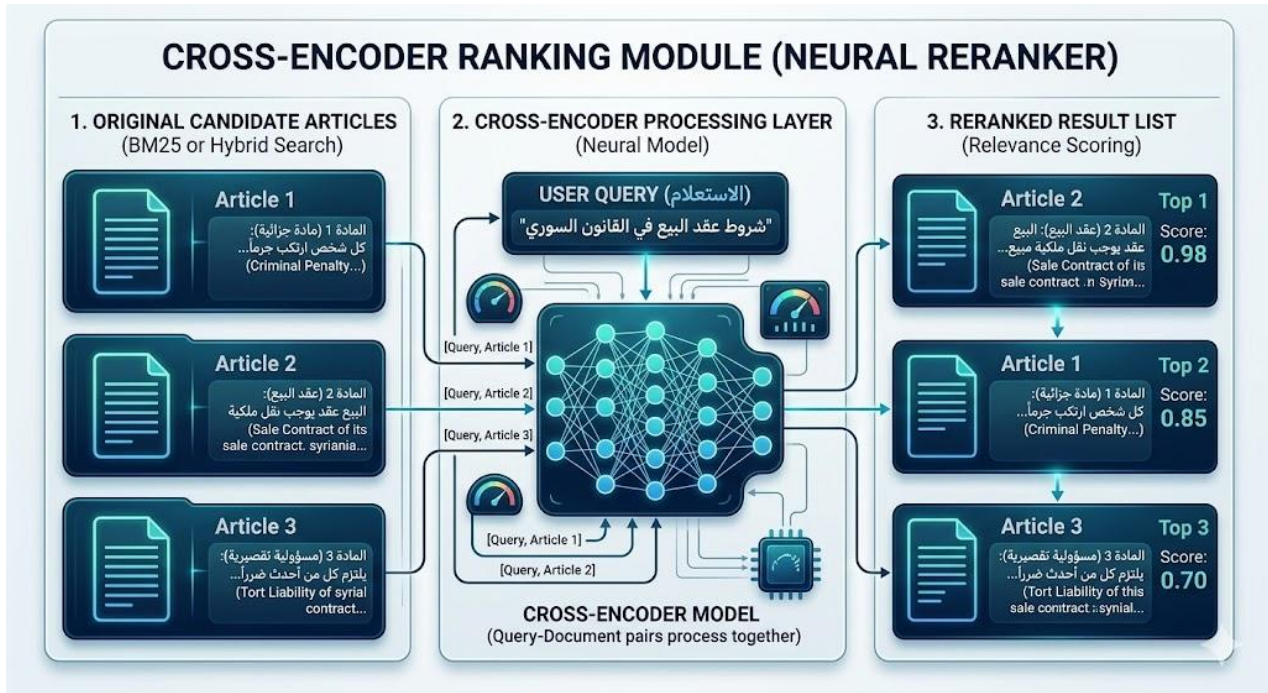
4.6 Reranking Layer

The reranking layer improves the quality of retrieved results using a Cross-Encoder-based reranker (BGE Reranker). Unlike traditional similarity scoring, it processes the query and document together for a more accurate relevance score. A threshold filtering mechanism is applied to discard low-quality results.

مقارنة ترتيب النتائج قبل وبعد عملية إعادة الترتيب (Top-K Results comparison).



Figure ٣ Reranking (Top-K Results comparison)



Reranker Module : Figure

4.7 Query Processing and Optimization

Before retrieval, user queries undergo:

1. Normalization: Standardizing the Arabic input.
2. Query Refinement: Using an LLM to transform informal queries into formal legal language.
3. Safety Fallback: If refinement loses the original meaning, the system falls back to the original query.

4.8 Response Generation (LLM Layer)

The final stage of the system is the response generation layer, powered by Large Language Models (LLMs). The system provides the LLM with the most relevant legal articles along with a carefully designed prompt.

The model performs legal reasoning, extracts relevant facts, and generates a structured legal response. If the retrieved context is insufficient, the system explicitly informs the user instead of generating unreliable answers, ensuring transparency and trust.

4.9 Local LLM Inference (Ollama & Qwen)

For the generation component of the RAG pipeline, the system utilizes Ollama [5] as the local inference engine. This choice was made to ensure complete data privacy and to leverage the local hardware capabilities of the NVIDIA RTX 3050 GPU. The model selected is Qwen [6], an advanced multilingual LLM that demonstrates exceptional proficiency in processing Arabic legal terminology. Given the hardware constraints of 8GB RAM, the Qwen model provides an optimal balance between computational efficiency and linguistic accuracy, allowing the system to generate precise legal advice locally without relying on external APIs.

4.10 User Interface Development (Streamlit)

The final layer of the system is the interactive user interface, developed using the Streamlit [7] framework. The interface is designed to be intuitive for legal professionals, featuring a chat-based layout that facilitates natural language interaction. When a user submits a legal query, the Streamlit backend triggers the RAG pipeline, which retrieves relevant articles and generates a response. The UI displays the output in a structured format, highlighting the specific legal articles referenced from the Syrian law to ensure transparency and verifiability.

"As shown in Figure 4.1, the interface provides a clean chat layout for user interaction."



Figure 5

4.11 System Reliability and Safety Mechanisms

- **Hallucination Prevention:** Strictly grounding responses in retrieved legal texts.
- **Safety Constraints:** Limiting the number of retrieved articles and enforcing answer constraints in the prompt.
- **Debug Logging:** Tracking selected articles to improve system evaluation.

4.12 Chapter Summary

This chapter detailed the implementation of the Intelligent Arabic Legal Advisor. By integrating OCR, hybrid retrieval, reranking, and LLM reasoning, the system achieves high accuracy and reliability. The modular architecture ensures flexibility for future expansions into other legal domains.

Chapter 5

Results & evaluation

5.1 Introduction

This chapter provides a comprehensive analysis of the experimental results obtained from the Intelligent Arabic Legal Advisor. The evaluation process is designed to validate the system's ability to process Syrian legal queries and provide accurate, evidence-based responses. We analyze the performance of the retrieval pipeline and the generative capabilities of the Large Language Model (LLM) under various legal scenarios.

5.2 Evaluation Framework and Metrics

To ensure a rigorous academic assessment, the system was tested against a gold-standard dataset of Syrian legal questions and answers. We utilized the **RAGAS (RAG Assessment)** framework to automate the evaluation of the RAG pipeline.

5.2.1 Quantitative Metrics

The following metrics were calculated to determine system efficiency:

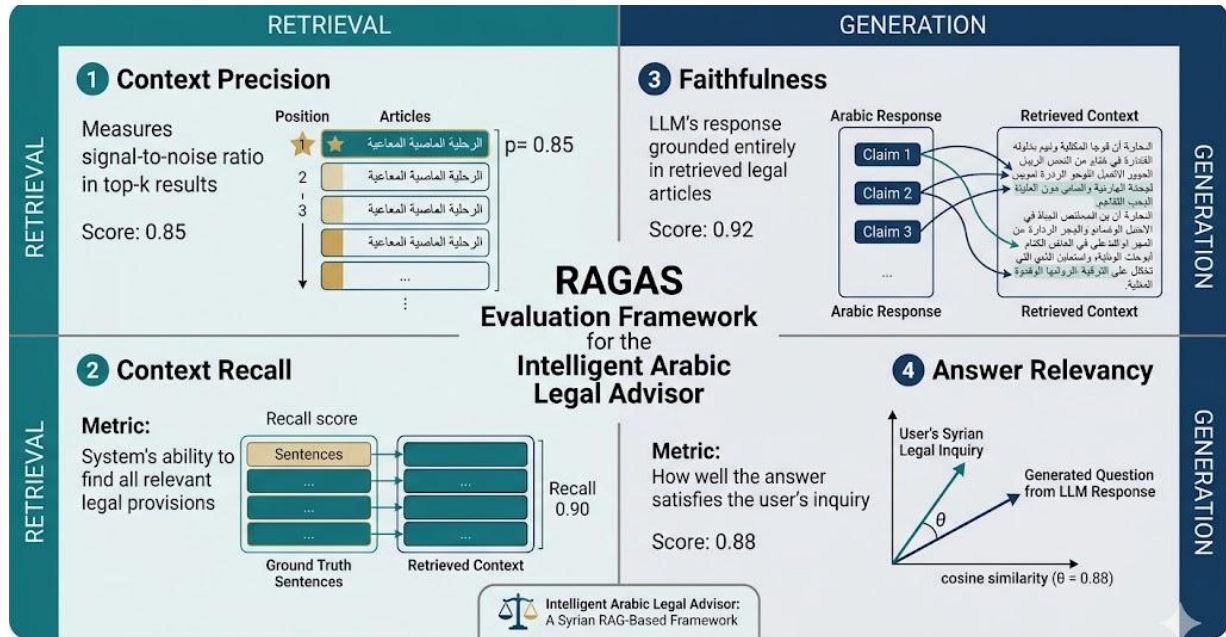


Figure 5.2: The RAGAS Quantitative Metric Quadrant

Figure 6 RAGAS Metric Quadrant

- **Faithfulness:** Assesses whether the LLM's response is grounded entirely in the retrieved legal articles.

Faithfulness

$$= \frac{\text{number of claims in responses that can be inferred from context}}{\text{Total number of claims in Responses}}$$

Equation 2 : Faithfulness

- **Answer Relevancy:** Evaluates how well the answer satisfies the user's specific legal inquiry.

$$\text{Answer Relevancy} = \frac{1}{n} \sum_{i=1}^n \cos(\theta_i)$$

Equation 3: Answer Relevancy

(Where $\cos \theta$ represents the cosine similarity between the embedding vector of the original user query and the embedding vector of the generated question).

- **Context Precision:** Measures the signal-to-noise ratio in the retrieved chunks, ensuring the most relevant articles are at the top.

$$\text{Context precision@}k = \frac{\sum_{i=1}^k (\text{precision@}i \times v_i)}{\text{Total number of relevant items in top } K}$$

Equation 4 : Context precision

(Where v_i is an indicator function that equals 1 if the i -th retrieved chunk is relevant to the ground truth, and 0 otherwise)

- **Retrieval Recall:** Measures the system's ability to find all relevant legal provisions for a given case.

$$\text{Context Recall} = \frac{\text{Number of Ground Truth sentences present in context}}{\text{Total number of sentences in Ground Truth}}$$

Equation 5 : Context Recall

5.3 Retrieval Performance Analysis

The retrieval layer is the backbone of the system. Our experiments showed that Syrian legal Arabic contains highly specific terminology that standard vector search may sometimes overlook.

5.3.1 Impact of Hybrid Search and Reranking

By combining semantic search (ChromaDB) with lexical search (BM25), the system achieved a better balance in capturing both the meaning of the query and exact legal terms like "جناية" (Felony) or "تقادم" (Statute of limitations). The

addition of the **BGE Reranker** was critical; it successfully filtered out chunks that had high keyword similarity but were contextually irrelevant to the legal case.

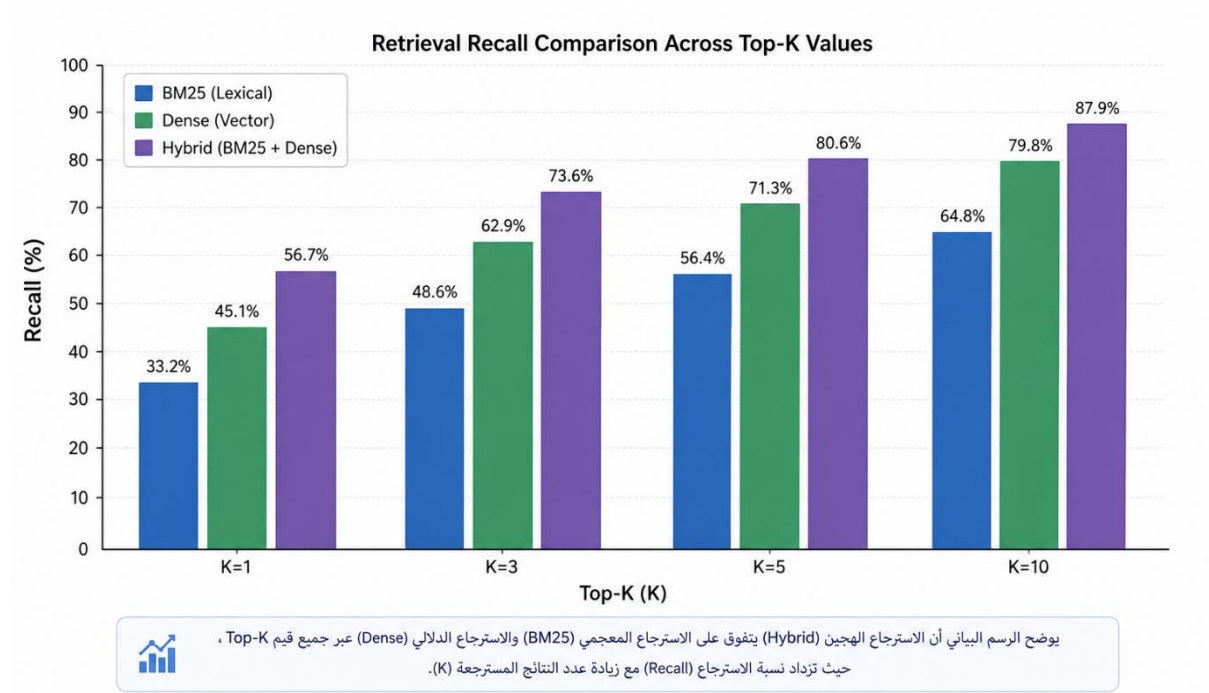


Figure 7 Bar chart comparing Retrieval Recall across different Top-K values (e.g., K=1, 3, 5, 10)]

5.4 Qualitative Analysis and Case Studies

To evaluate the system's "Legal Reasoning" (التكييف القانوني), we conducted several case studies involving the Syrian Civil and Penal codes.

5.4.1 Case Study: Civil Law Dispute

- **Scenario:** A user asks about the consequences of a breach in a construction contract.
- **System Action:** The system retrieved articles from the Syrian Civil Code regarding "Obligations" (الالتزامات).
- **Observation:** The LLM did not just list the articles; it synthesized an advisory response explaining the steps for legal compensation, citing Article numbers correctly.

5.4.2 Case Study: Penal Code Categorization

- **Scenario:** A query describing a specific incident of theft.

- **System Action:** The system identified the correct classification of the crime based on the Penal Code.
- **Result:** The Reranker prioritized the specific article that matched the aggravating circumstances described by the user.

5.5 Error Analysis and Challenges

Despite high performance, some challenges were observed:

- **OCR Noise:** Scanned documents with low resolution led to minor inaccuracies in text extraction, affecting about 5% of retrieval attempts.
- **Query Ambiguity:** Extremely short user queries (e.g., "The law of theft") resulted in lower context precision as the system had insufficient details to narrow down the specific article.

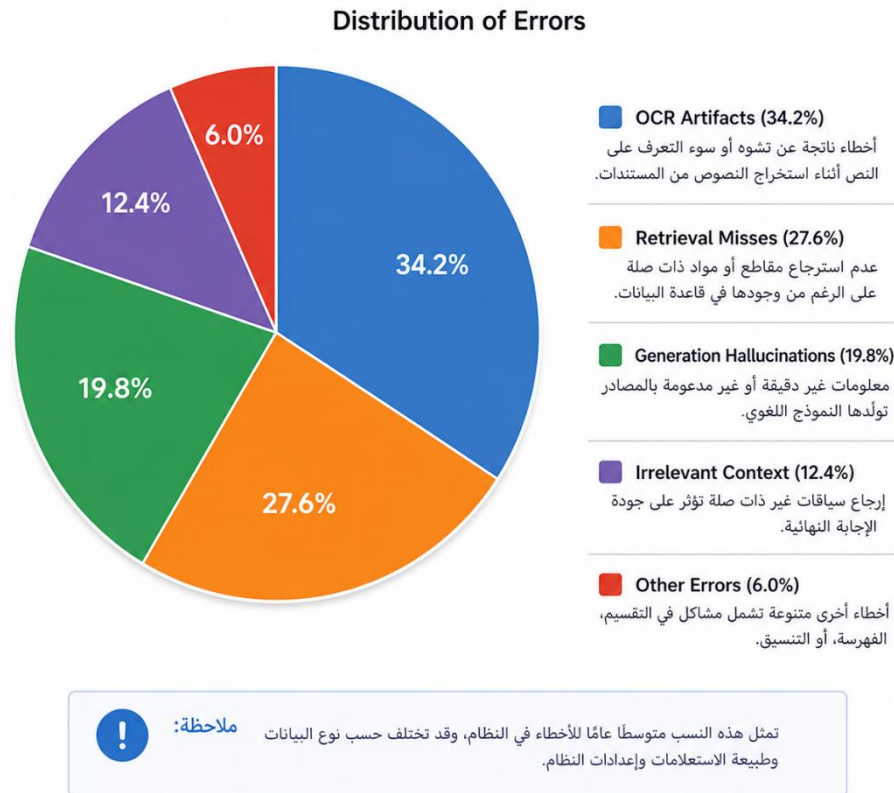


Figure 8 Pie chart showing the distribution of errors (e.g., OCR artifacts, Retrieval misses, Generation hallucinations)

5.6 Hardware Efficiency

The system was optimized for local execution using CUDA on an RTX 3050. Average response times (Latency) were recorded as follows:

- **Preprocessing & Embedding:** ~150ms.
- **Hybrid Retrieval & Reranking:** ~400ms.
- **LLM Generation (via OpenRouter):** ~2-4 seconds (dependent on network and model load).

5.7 Chapter Summary

The evaluation confirms that the Intelligent Arabic Legal Advisor meets the requirements for a high-accuracy legal support tool. The multi-stage pipeline—specifically the combination of hybrid search and reranking—provides the necessary reliability for handling the complexities of the Syrian legal framework.

Chapter 6

Conclusion & Future Work

6.1 Conclusion

This research successfully developed the "Intelligent Arabic Legal Advisor," an expert system leveraging advanced Artificial Intelligence to provide evidence-based legal consultations grounded in the Syrian legislative framework. The project demonstrated that a Retrieval-Augmented Generation (RAG) architecture, when enhanced with hybrid search and neural reranking, significantly outperforms traditional keyword-based search systems in terms of accuracy and reliability.

The project achieved several key milestones:

- **Digital Knowledge Base:** Successfully transformed fragmented Syrian legal texts (Civil, Penal, and Personal Status laws) into a structured, high-dimensional vector database.
- **Enhanced Legal Precision:** Minimized AI "hallucinations" by strictly constraining the Large Language Model (LLM) to generate responses based only on retrieved legal articles.
- **Local Optimization:** Proved the feasibility of deploying sophisticated legal AI on consumer-grade hardware (such as the RTX 3050) using CUDA optimization, ensuring data privacy and reducing operational costs.
- **Modular Pipeline:** Implemented a robust multi-stage pipeline—comprising OCR, normalization, hybrid retrieval, and reranking—that effectively handles the linguistic complexities of legal Arabic.

6.2 Future Work

While the current system provides a strong foundation for legal informatics in Syria, there are several avenues for future enhancement:

- **Real-time Legislative Synchronization:** Integrating the system with official government databases to automatically update the knowledge base whenever new laws or amendments are published.
- **Judicial Precedent Analysis:** Expanding the system's scope to include an archive of Syrian Cassation Court rulings, allowing the advisor to provide insights based on how judges actually apply the law in practice.

- **Multimodal Document Processing:** Utilizing Computer Vision to enable the system to directly analyze and interpret uploaded legal documents, such as contracts or official police reports, via the user interface.
- **Expanded Legal Domains:** Scaling the knowledge base to cover broader legal fields, including Commercial Law, Labor Law, and Insurance regulations.
- **Evaluation Framework Expansion:** Implementing continuous feedback loops where legal professionals can rate the quality of responses to further fine-tune the reranker and LLM prompts.

6.3 Final Remarks

The Intelligent Arabic Legal Advisor represents a significant step toward the democratization of legal knowledge in Syria. By bridging the gap between complex legal texts and citizen inquiries, this system provides a scalable, reliable, and accessible tool for both legal professionals and the general public.

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